

Engineering Design & Problem Solving (InSPIRESS)

TEKS §127.785 • Advanced • Senior • Mr. Gavaldon • Da Vinci School

Week 1

MONDAY

Day 1 - Procedures, Syllabus & Design Notebook

Aug 10

OBJECTIVE

Students will set up a professional engineering design notebook to documentation standards (numbered, dated, permanent, witness-ready), sequence the seven stages of the formal engineering design process, distinguish design criteria from constraints, and write a first notebook entry that frames a candidate InSPIRESS payload as a measurable design problem.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - professional standards & the engineering design notebook

ELPS / EB SUPPORT

ELPS/EB supports: pre-teach *criteria*, *constraint*, *iteration*, *specification* with a bilingual visual glossary and icons (PV, DP, VC); model the notebook setup live under the doc cam with numbered steps (CD, RS); provide sentence stems for the framed-problem entry — 'A successful design must ___, measured by ___ (units)'; 'A constraint I cannot violate is ___' (RS); pair EB students to talk through the criteria/constraint sort before writing (PN, WT); allow extra time on the notebook build (ET).

AGENDA • 45 MIN

6' Bell-work: In your notebook, write one sentence on what separates *engineering* from *building*. We'll test it against the professional definition.

10' The engineering design process: seven stages, and why documentation is 'stage zero'

10' The language of a real problem: criteria vs. constraints, framed around the InSPIRESS payload challenge

13' Build your design notebook to professional standard + write your first framed-problem entry

6' Exit ticket: sort four statements into criteria/constraint and write one measurable success criterion

SLIDES / ACTIVITY

Engineering Design & the Professional Notebook

NOTEBOOK

Set it up today: cover, table of contents, numbered/dated pages. First entry = one thing you want to design or present this year.

TURN IN

Design Notebook Standard + Define-Phase Problem Framing

OBJECTIVE

Students will analyze technical communication using the claim–evidence–interpretation structure, apply significant-figure rules to measured quantities, convert and combine units by dimensional analysis with cancellation shown, and report a measured result with appropriate uncertainty.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - technical communication & presentation; applied measurement (units/dimensional analysis)

ELPS / EB SUPPORT

ELPS/EB supports: the video has a guided viewing sheet with sentence stems and is captioned (RS, CD, VC); pre-teach *significant figures*, *dimensional analysis*, *uncertainty*, *precision/accuracy* with a bilingual glossary and a kept conversion table (PV, DP); the calculation is modeled I-do / we-do / you-do with units written on every line (CD, RS); pair EB students for the you-do (PN, WT); extra time on the problem set (ET).

AGENDA • 45 MIN

5' Bell-work: Write the best presentation you've ever seen and one concrete reason it worked.

8' Watch a working engineer present, then name the *claim → evidence → interpretation* moves she makes

7' The unit of a technical argument: claim → evidence → interpretation

13' Significant figures, units & dimensional analysis — I do / we do / you do

8' Measurement uncertainty: precision vs. accuracy and how to report it

4' Exit ticket

SLIDES / ACTIVITY

Present Like an Engineer: Evidence, Units & Uncertainty

NOTEBOOK

Write the 3 presentation rules. Keep the unit-conversion example with all steps shown.

TURN IN

Technical Communication + Measurement: Evidence, Sig Figs, Units, Uncertainty

OBJECTIVE

Students will apply lab and field safety protocols — PPE selection, hazard identification, and a severity × likelihood risk assessment — and conduct a structured measurement investigation, computing the mean and range of repeated trials and naming the sources of measurement uncertainty.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - safe, responsible lab & field investigation practices

ELPS / EB SUPPORT

ELPS/EB supports: safety rules shown with icons and a live demo (VC, DP); pre-teach *hazard*, *risk*, *PPE*, *SDS*, *control* with a bilingual glossary (PV); the risk matrix and the mean/range calculation are modeled step-by-step (CD, RS); investigation done in pairs so EB students verbalize procedure and data (PN, WT); extra time to complete the log (ET).

AGENDA • 45 MIN

5' Bell-work: name one hazard in this room and one control that reduces its risk.

10' Safety as engineering: hazard vs. risk, the hierarchy of controls, and a risk-assessment matrix

6' How engineers investigate: measurable question, controlled variables, repeated trials

16' Structured investigation: measure an object 5× and analyze the spread

5' Precision vs. accuracy: interpret your data + name error sources

3' Sign the safety agreement + exit

SLIDES / ACTIVITY

Safe Investigation: Hazards, Risk & Real Data

NOTEBOOK

Glue in the safety agreement. Record your investigation measurements + what you noticed.

TURN IN

Risk Assessment (JHA) + Measurement Investigation Log

OBJECTIVE

Students will form engineering teams and assign accountable roles with a RACI chart, use a weighted decision matrix to select among candidate payload concepts, and deliver a 90-second evidence-based team pitch structured as claim → evidence → interpretation.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - teamwork & collaboration; short-form presentation

ELPS / EB SUPPORT

ELPS/EB supports: pre-teach *RACI*, *decision matrix*, *weight*, *traceability* with a bilingual glossary and a worked example on the board (PV, DP, VC); the pitch uses a claim–evidence–interpretation sentence-frame planner (CD, RS); teams rehearse together before presenting so EB students practice first (PN, WT); allow notes during the pitch and extra time to plan (ET).

AGENDA • 45 MIN

5' Bell-work: describe one team you were on that worked — what made it work?

9' Engineering team roles, RACI accountability, and the shared notebook

8' Choosing objectively: the weighted decision matrix

13' Plan a 90-second pitch: claim → evidence → interpretation

8' Deliver mini-pitches + peer scoring with the rubric

2' Exit ticket

SLIDES / ACTIVITY

Engineering Teams, Objective Decisions & the First Pitch

NOTEBOOK

Write your team, your role, and your 60-second pitch outline.

TURN IN

Team Charter + Concept-Selection Decision Matrix + Pitch Plan

OBJECTIVE

Students will demonstrate mastery of the week's foundations — the design process, criteria vs. constraints, significant figures and unit conversion, measurement uncertainty, risk assessment, and the weighted decision matrix — by solving application problems in a team review, and will create and verify an Autodesk Education account to enable Fusion 360.

TEKS

§Eng Design & Presentation TEKS (draft - verify official section) - review of foundations; digital citizenship & accounts

ELPS / EB SUPPORT

ELPS/EB supports: the review is team-based and low-stakes, with wait time before answers (PN, WT); account steps are numbered and screenshotted in Classroom with a bilingual walkthrough (VC, CD, DP); pre-teach *parametric*, *verify*, *credential* (PV); rephrase problems on request and allow extra time (RS, ET).

AGENDA • 45 MIN

5' Bell-work: two quick recalls + one calculation (units + safety)

16' Team review challenge: application problems across the week's skills

16' Create & verify your Autodesk Education account (Fusion 360)

5' Account check + info card + digital-citizenship reminder

3' Exit ticket + preview next week (parametric sketching)

SLIDES / ACTIVITY

Prove It: Week-1 Synthesis + Get CAD-Ready

NOTEBOOK

Note your Autodesk account email (not your password) and one thing you want to model first.

TURN IN

Week 1 Foundations — Cumulative Challenge, Autodesk Account — Info Card + Digital Citizenship